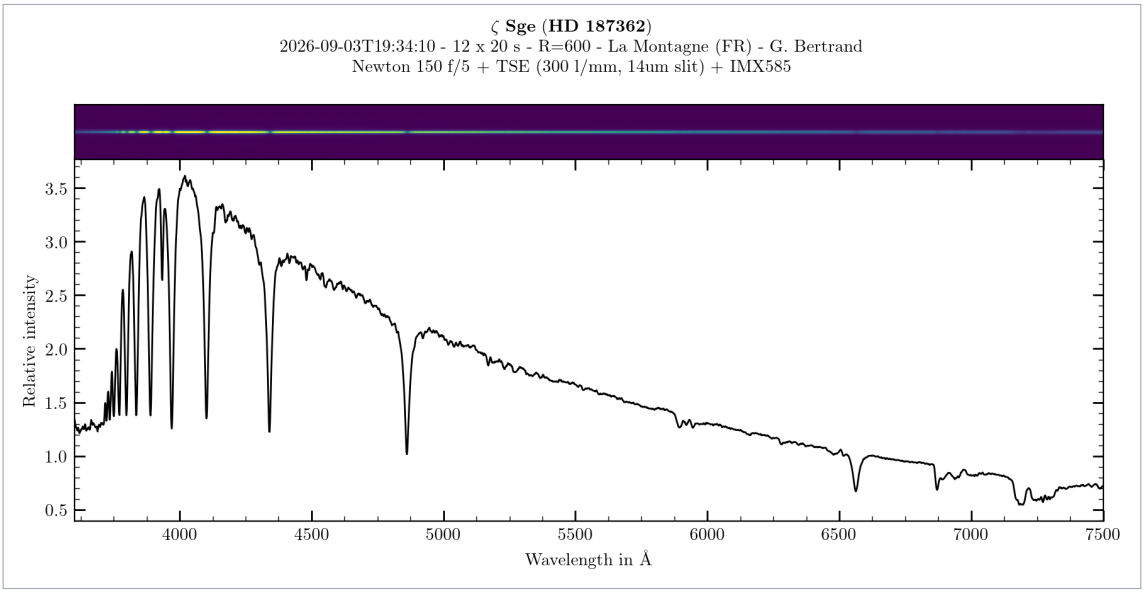
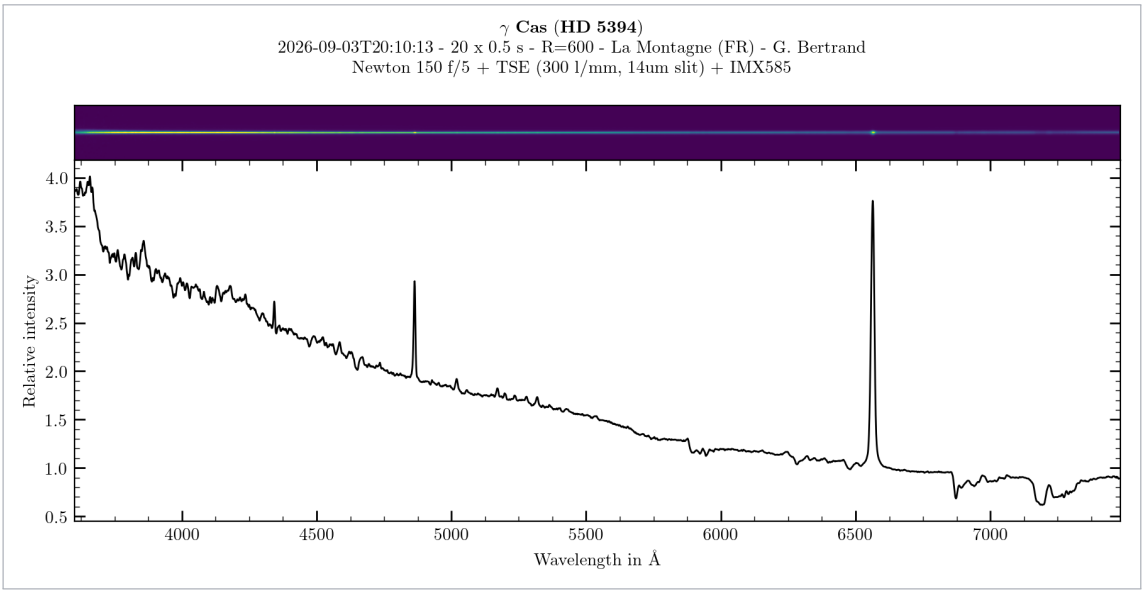
TinyStar’Ex

Examples of spectra obtained with TinyStar’Ex

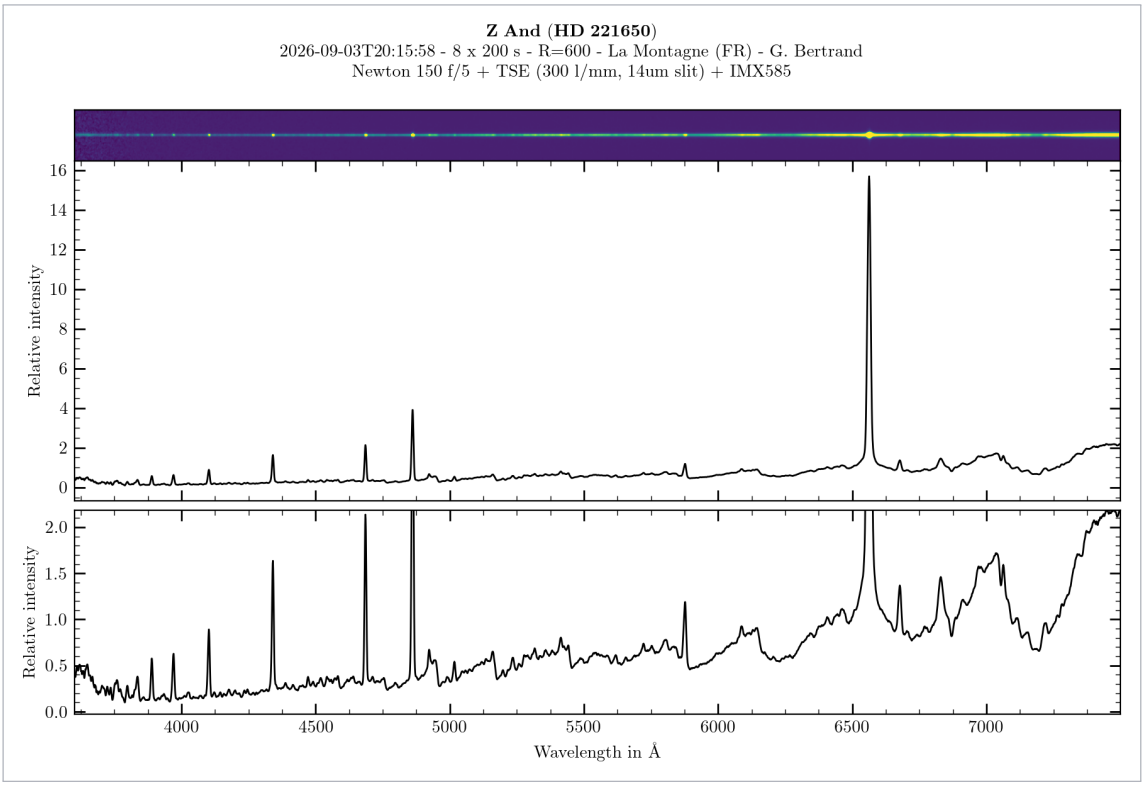
Spectrum of the star ζ Sge. TSE on a 150 mm f/5 Newtonian telescope. Guillaume Bertrand.



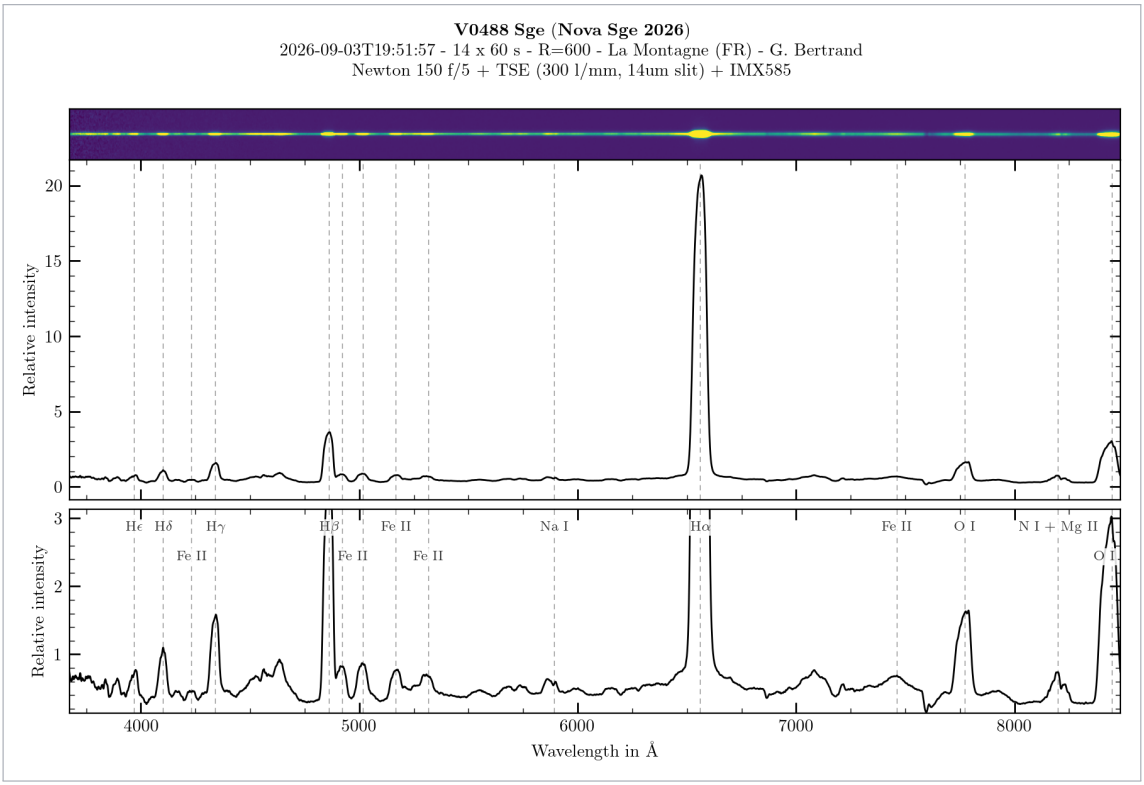
Spectrum of the Be star γ Cas. TSE on a 150 mm f/5 Newtonian telescope. Guillaume Bertrand.



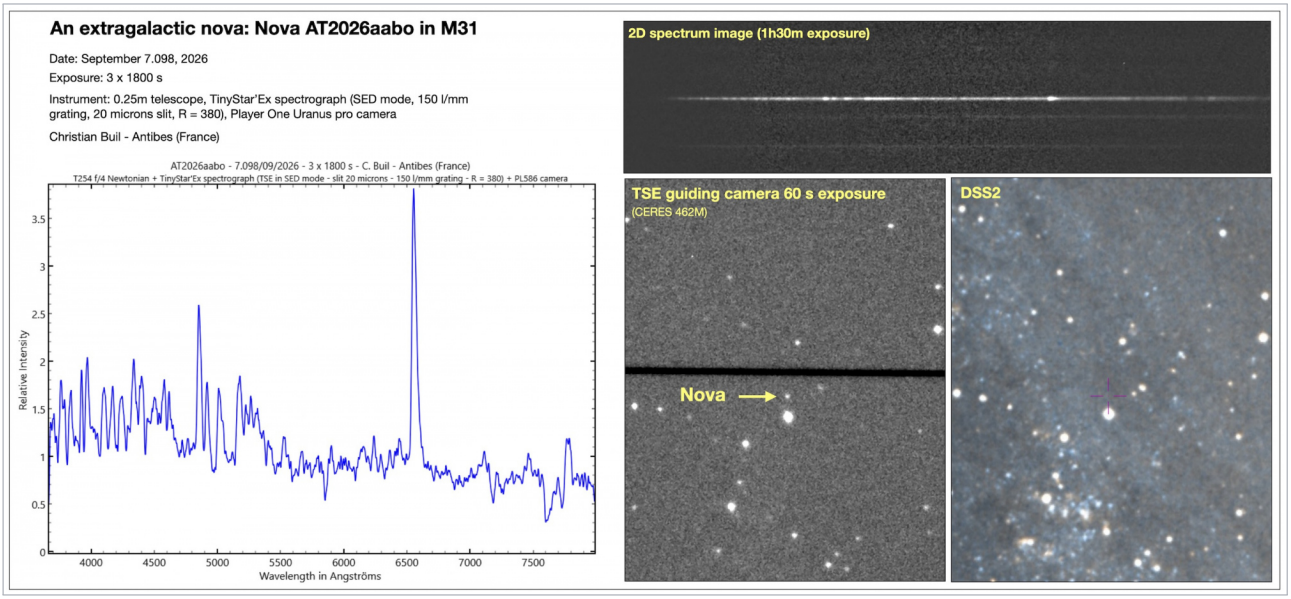
Spectrum of the symbiotic star Z And. TSE on a 150 mm f/5 Newtonian telescope. Guillaume Bertrand.



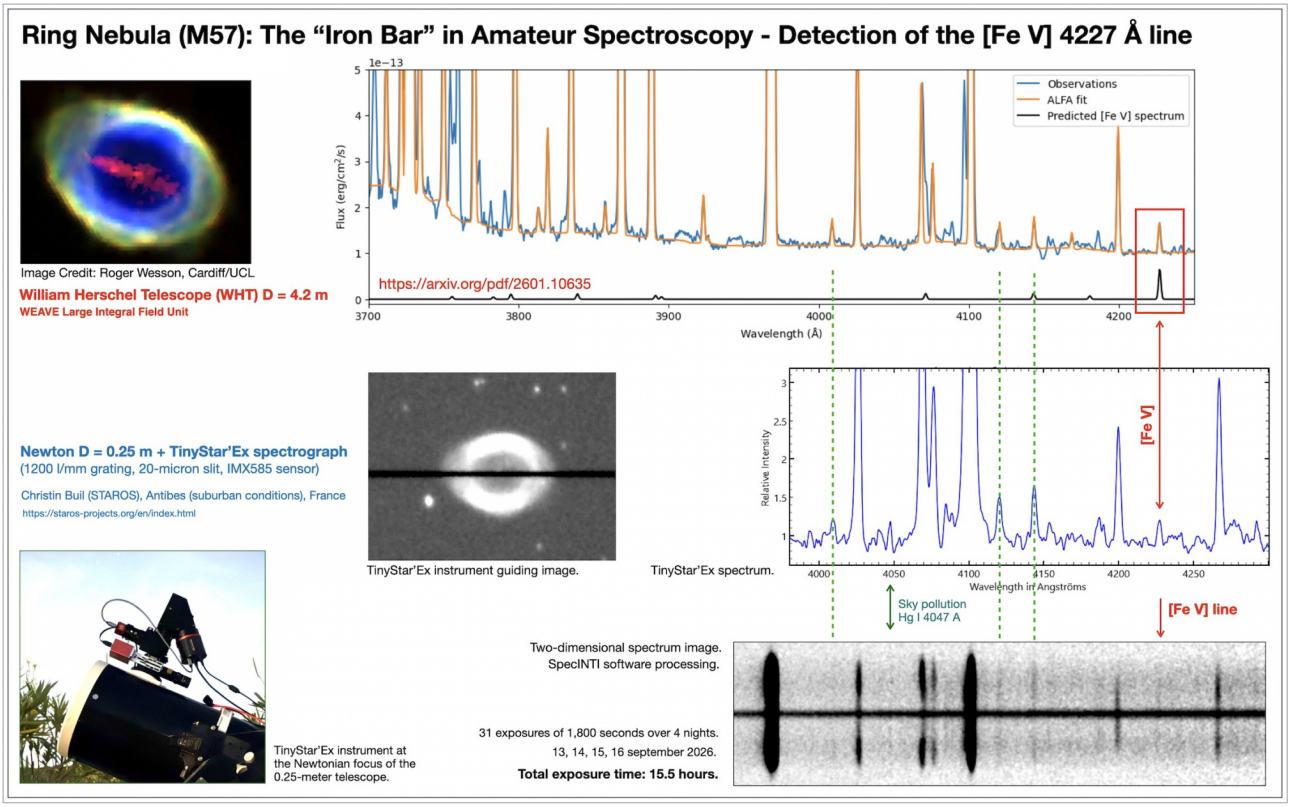
Spectrum of the nova V488 Sge. TSE on a 150 mm f/5 Newtonian telescope. Guillaume Bertrand.



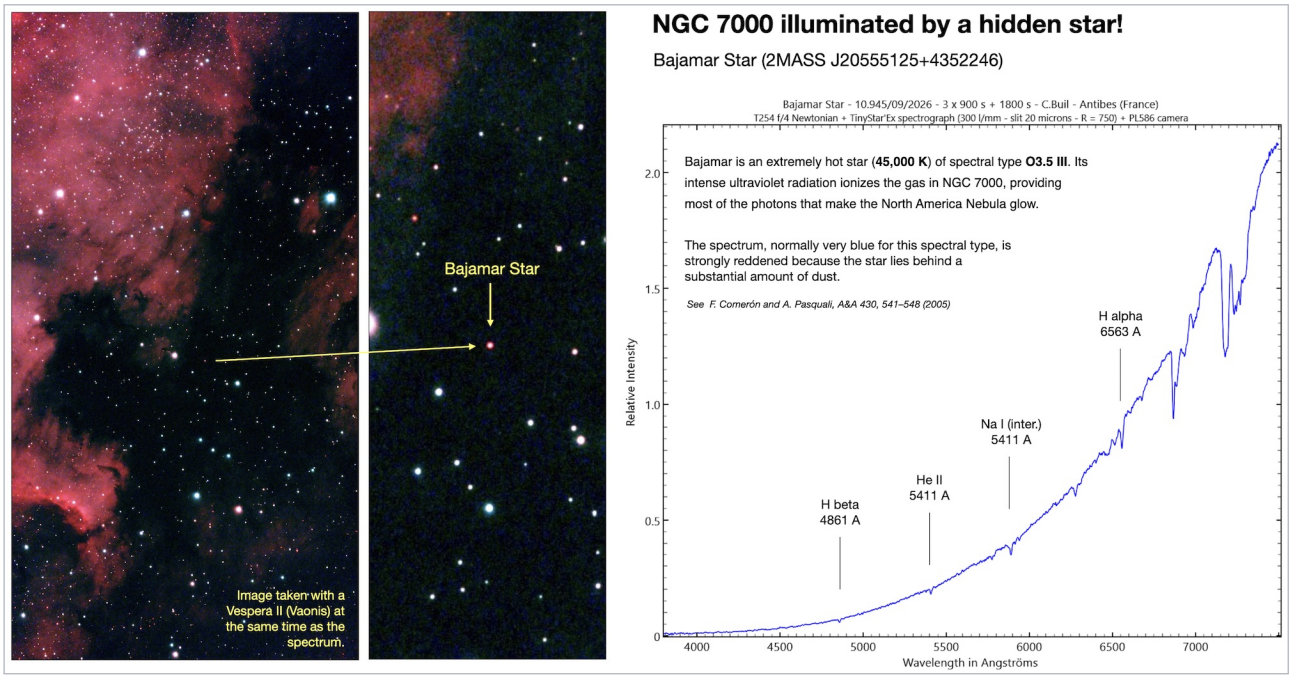
Spectrum of the extragalactic nova AT2026aabo (in Messier 31). TSE on a 250 mm f/4 Newtonian telescope. Christian Buil.



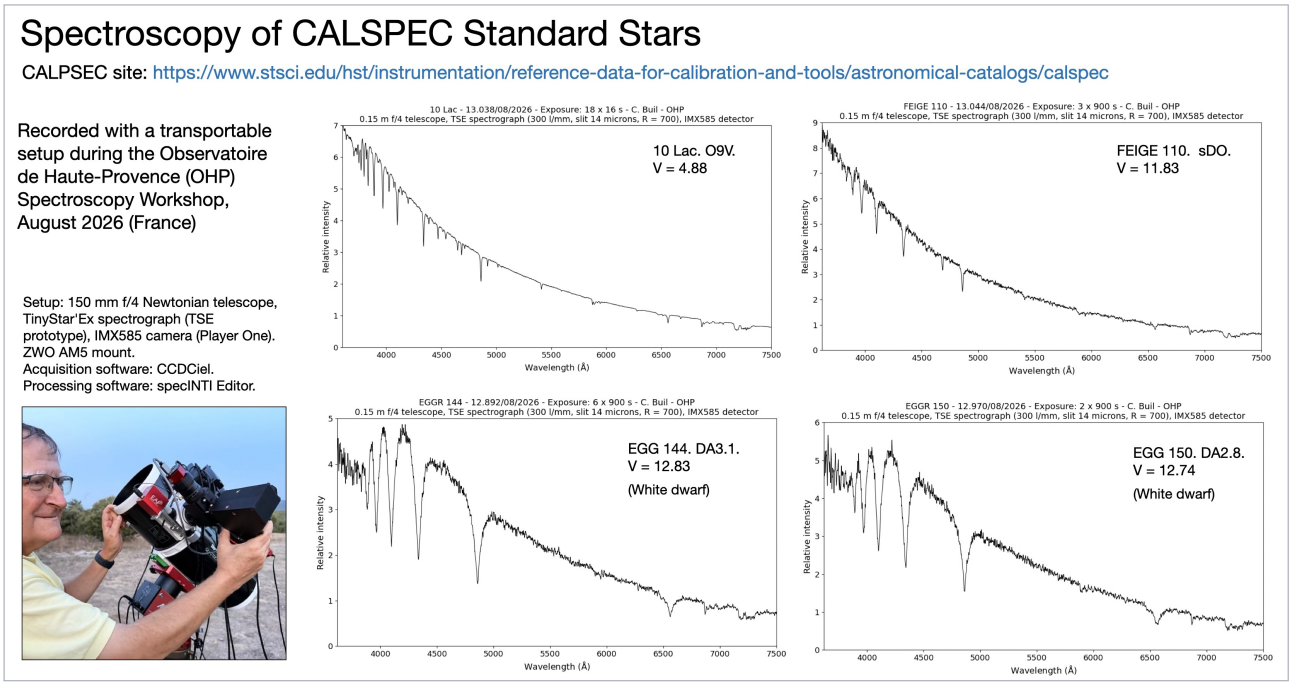
Spectrum of the recently discovered iron bar in Messier 57 (long exposure!). TSE on a 250 mm f/4 Newtonian telescope. Christian Buil.



Spectrum of the hidden star that illuminates the large nebula NGC 7000 (North America Nebula). TSE on a 250 mm f/4 Newtonian telescope. Christian Buil.

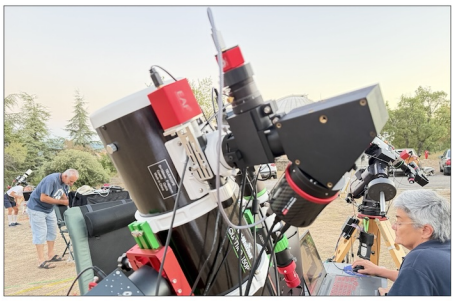


Spectra of spectrophotometric standards (very hot stars and white dwarfs). TSE on a 150 mm f/4 Newtonian telescope. Christian Buil.

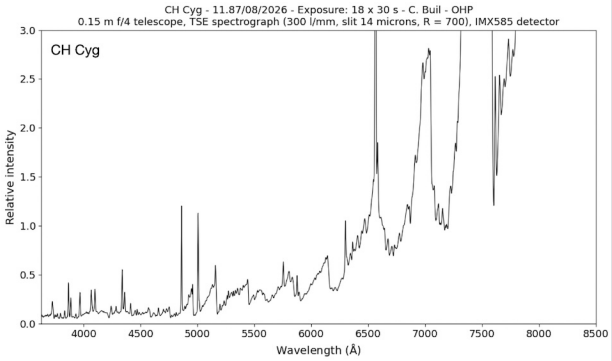


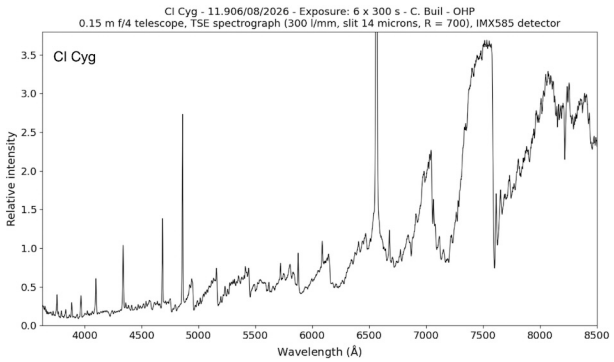
Spectra of symbiotic stars with a very compact telescope. TSE on a 150 mm f/4 Newtonian telescope. Christian Buil.

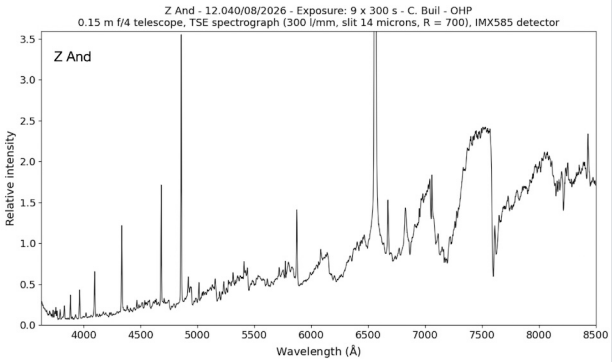
Observation of symbiotic stars during the 2026 spectroscopy workshop at the Observatoire de Haute-Provence (France). STAROS association



The D=150 mm f/4 telescope equipped with a TinyStar'Ex spectrograph. The equatorial is a ZWO AM5. Acquisition: CCDCiel. Processing: specINTI Editor.



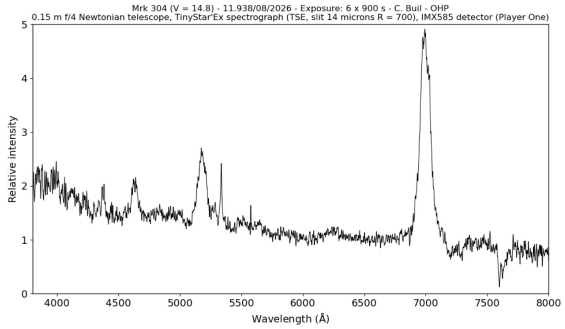




Spectrum of the galaxy Markarian 304 as a function of telescope aperture and observing site (with and without light pollution). TinyStar'Ex spectra. Christian Buil.

Observation of the galaxy Markarian 304 (Seyfert type 1 galaxy, V = 14.7) - Comparison between a professional observatory site and a suburban site

Note: the spectrograph and exposure time are the same, but not the telescope (0.150 m vs 0.254 m)





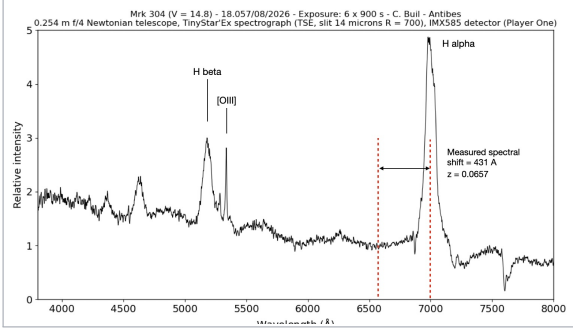
Observatoire de Haute-Provence (OHP)

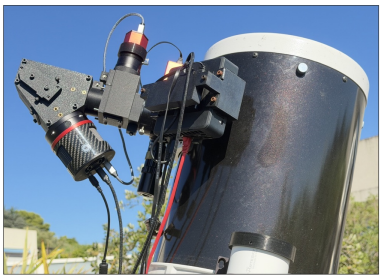
OHP 2026 Spectroscopy Workshop (STAROS)

0.15 m f/4 Newtonian telescope - T**inyStar'Ex** (TSE) spectrograph - IMX585 detector.

14-micron slit (4.8 arcseconds on sky).
Exposure: 6 x 900 seconds

Resolving power at 6563 Å (R): **700**
Signal measured at 6563 Å: **3600 ADU**
Measured signal-to-noise ratio: **17**





Observatoire Antibes Saint-Jean

Suburban conditions

0.254 m f/4 Newtonian telescope - T**inyStar'Ex** (TSE) spectrograph - IMX585 detector.

14-micron slit (2.8 arcseconds on sky).
Exposure: 6 x 900 seconds

Resolving power at 6563 Å (R): **700**
Signal measured at 6563 Å: **13500 ADU**
Measured signal-to-noise ratio: **36**

G = 0.895 e-/ADU, RON = 1.02 e-, Dark = 0.0019 e-/s